

Capita Selecta – Project Idea

Comparing Li-Fi, Bluetooth, and Wi-fi for Sending Sensor Data

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Abbreviations

Term	Meaning
Li-Fi	A method that sends data through light, using a LED as the transmitter and a photodiode as the receiver
Wi-Fi	A common wireless method that sends data through radio waves, often used for internet access
BLE	Bluetooth Low Energy, a short-range wireless method that sends data through radio waves and uses little power
Raspberry Pi Pico 2 W	A small, low-cost microcontroller board with built in Wi-Fi and Bluetooth
MQTT	A lightweight messaging method used to send data from one device to another, often used for sensor data
KMP	Kotlin Multiplatform, a way to write shared code once and run it on several platforms, such as Android, iOS, web, and desktop
OOK	On Off Keying, a way to send data by turning a signal, such as light, on and off
BER	Bit Error Rate, a measure of how many bits are received wrong compared to how many were sent
INA219	A small sensor chip that measures voltage and current, used here to measure power use
Internal temperature sensor	A sensor built into the Raspberry Pi Pico 2 W chip itself, which measures the chip's own internal heat. It rises when the chip works harder, for example while sending data over Wi-Fi, BLE, or Li-Fi, so it works as a sign of how much load each method places on the chip
Native	In this document, a native chart or a native notification means a tool built specifically for one platform, such as Android or iOS, rather than a shared tool used across all platforms. Each platform draws its own charts and shows its own notifications with its own separate library
EC	European Credit, a unit that shows how many study hours a course requires
GPIO	General Purpose Input/Output
PWM	Pulse Width Modulation

Table of Contents

1. Project Overview.....	4
2. Assignment Description	5
1.1. <i>Argumentation.....</i>	<i>6</i>
1.2. <i>Research question</i>	<i>7</i>
1.3. <i>Application and relevance.....</i>	<i>7</i>
3. Supervisor.....	7
4. Duration.....	7
5. Deliverables	7
6. Estimated Components	8
7. Estimated Hour Breakdown.....	9

1. Project Overview

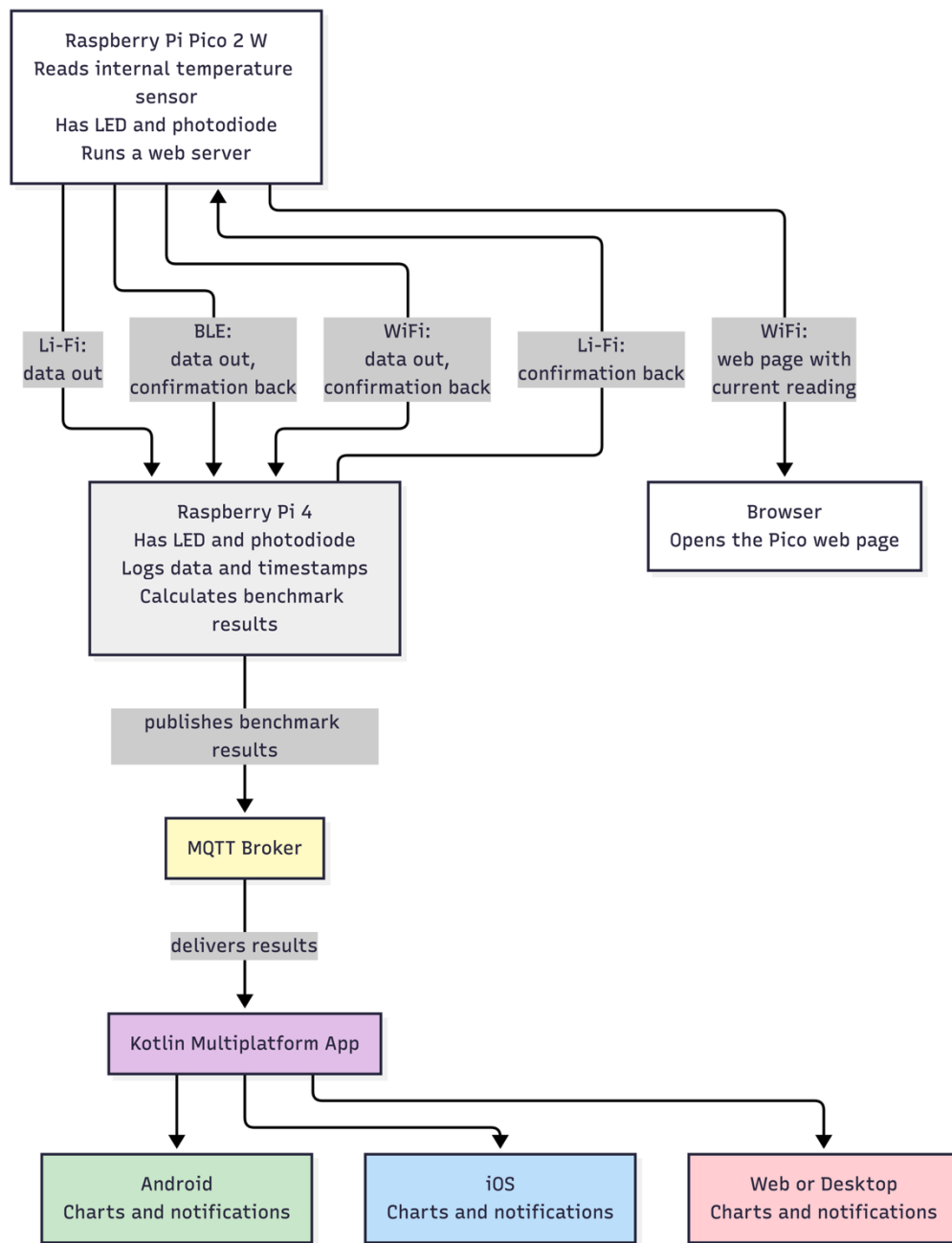


Figure 1: Overview of the sensor data path from the Raspberry Pi Pico 2 W to the Kotlin Multiplatform app

Figure 1 depicts a Raspberry Pi Pico 2 W reads its internal temperature sensor, then sends the same value through three separate paths at the same time. The Raspberry Pi Pico 2 W and Raspberry Pi 4 can send and receive light signals, so Li-Fi flows both ways. The Raspberry Pi 4 and the Raspberry Pi Pico 2 W both use their LEDs and photodiodes to send and receive

confirmation signals as light. The Bluetooth Low Energy (BLE) path and the Wi-Fi path also move in both directions, since the Raspberry Pi Pico 2 W sends its data straight to the Raspberry Pi 4, and the Raspberry Pi 4 sends a confirmation back over the same channel. The three methods carry their own confirmation, so the comparison between Li-Fi, BLE, and Wi-Fi stays complete on both the forward path and the return path. The Raspberry Pi 4 logs data, timestamps incoming signals, and calculates benchmark results such as throughput, reliability, latency, and power use. After that, the Raspberry Pi 4 sends these results out through MQTT. The Kotlin Multiplatform app then receives this feed and shows the live data and benchmark results on each platform. Each platform also makes its own charts and displays its own notifications. The Raspberry Pi Pico 2 W also runs a small web server of its own, reachable over Wi-Fi from a phone browser, which shows the current reading directly from the device itself. This gives a second, independent way to check that the value shown in the app matches the value the Raspberry Pi Pico 2 W holds at that moment, which helps verify that the whole system reports data correctly in real time.

2. Assignment Description

This project builds and tests three different wireless methods that send sensor data from an Raspberry Pi Pico 2 W to a Raspberry Pi 4. The three methods are Li-Fi, which uses a LED and a photodiode, Bluetooth Low Energy, and Wi-Fi. The same sensor readings are sent through all three methods at the same time. This allows a direct comparison of throughput, range, reliability measured as bit error rate, latency, and power use. The comparison also covers how each method behaves under interference, since Wi-Fi and BLE share the same radio spectrum, while Li-Fi depends on a clear path of light between transmitter and receiver.

The sensor value used for this comparison is the Raspberry Pi Pico 2 W's own internal temperature sensor. This value rises with processing load and radio activity, so it can work as a supporting indicator of device stress during transmission. Since a single Raspberry Pi Pico 2 W sends the reading through Li-Fi, BLE, and Wi-Fi at the same time for the main comparison, a temperature rise during that combined test cannot be tied to one method alone, since all three are active together. For this reason, the temperature and power comparison is run separately, with each method tested on its own in a dedicated window while the other two stay idle, so any temperature change during that window can be attributed to that one method and checked against its matching INA219 power reading.

Each reading also receives a confirmation once it arrives at the Raspberry Pi 4, and this confirmation travels back through the same method that carried the original reading. For BLE and Wi-Fi, this is straightforward, since both methods already support two-way communication. For Li-Fi, the Raspberry Pi 4 has its own LED, and the Raspberry Pi Pico 2 W has its own photodiode circuit, so the confirmation travels back as light as well, rather than switching to a different medium. This means Li-Fi is tested as a complete, working round trip, not only as a one-way data path, which gives a honest picture of what a light-based connection can and cannot do, including any extra delay or failure that a return flight path adds on its own.

The Raspberry Pi 4 receives raw data from all three transmission paths using the same simple transport method, so the comparison stays fair. The Pi4 timestamps the data, logs it, and calculates the benchmark results. Once this measurement stage is complete, the Pi4 publishes the results through MQTT. A Kotlin Multiplatform application then shows the live sensor data and the benchmark results on several platforms. The application shares its data models, its MQTT client, its benchmark logic, and its state handling through Kotlin Multiplatform, but each platform uses its own chart library and its own notification system.

The Raspberry Pi Pico 2 W also hosts a small web server, reachable from a phone browser over Wi-Fi, which shows the current reading directly from the device. This does not need any extra hardware, since the Raspberry Pi Pico 2 W's Wi-Fi radio is already used for one of the three transmission methods. This gives an independent way to verify the value shown in the Kotlin Multiplatform app, since the value on the device itself and the value shown in the app can be checked against each other in real time, confirming that the full path from sensor to app reports the correct value rather than a delayed or incorrect one.

1.1. Argumentation

Wi-Fi and BLE are common choices for sensor communication, so their behavior is well known. Li-Fi is far less common in student projects, yet it has real industry relevance in places that restrict radio signals, such as hospitals or aircraft, or in cases where a signal confined to light offers better protection against interference or interception.

This project does not rely on claims from papers alone, for it measures these differences directly and reports the real results. This gives the project depth on both sides of the study program.

On the electrical engineering side, the work includes analog circuit design for the photodiode receiver, firmware for encoding and decoding the optical signal, configuration of BLE and Wi-Fi radios, and a controlled test method that includes interference testing and repeated trials.

A single Raspberry Pi Pico 2 W handles all three methods at the transmitter side, since driving the Li-Fi LED only uses a plain GPIO or PWM pin, which is fully separate from the radio hardware. Wi-Fi and BLE, however, share the same physical radio on the chip, and the chip manages this through a coexistence mechanism, switching rapidly between the two. This is a standard, well supported setup, though it can add small timing variation between Wi-Fi and BLE, which the report states clearly as a known hardware characteristic rather than an unexplained result.

On the computer science side, the work includes a real architecture decision for Kotlin Multiplatform, a data pipeline that aggregates and benchmarks live data, and a working cross platform dashboard.

The internal temperature reading also needs a clear limit stated in the report, for the value is affected by CPU load and by room temperature at the same time, so a rise in temperature alone does not prove one single cause. It is treated as a supporting signal alongside the INA219 power readings, not as a standalone, precise measurement, and it is only compared per method

during the separate, isolated test runs described above, not during the combined, simultaneous test.

1.2. Research question

Under which conditions of distance, interference, and obstruction does each of Li-Fi, BLE, and Wi-Fi perform best for sending sensor data, based on throughput, range, reliability, latency, and power use?

1.3. Application and relevance

This system fits sensor deployment in places that restrict radio signals, such as hospitals, aircraft, or sensitive industrial sites. It also fits secure sensor links, since a light-based signal cannot leave a room without a visible path. In addition, it supports engineering decisions for IoT sensor networks, for the results are based on measured data rather than assumption. It also points toward a design pattern where a sensor sends data through more than one medium, so a signal blocked on one path may still arrive through another.

Li-Fi is not treated as a replacement for Wi-Fi. Its need for a clear light path and its shorter working range make it useful in specific situations rather than as a general solution. The report studies this limit directly rather than assuming it away.

3. Supervisor

The supervisor for this project is Mr. Christian Slot.

4. Duration

Half a year, which matches the usual length of a 10 EC module of 280 hours.

5. Deliverables

- A working sensor node that sends the same internal temperature reading through Li-Fi, BLE, and Wi-Fi at the same time
- A two-way Li-Fi link, with a LED and a photodiode circuit on the Raspberry Pi Pico 2 W side, and a matching LED and photodiode circuit on the Raspberry Pi 4 side, each photodiode circuit built with a transimpedance amplifier and a comparator stage, soldered on small boards for stable use
- A confirmation mechanism for all three methods, where the Raspberry Pi 4 sends a message back to the Raspberry Pi Pico 2 W once a reading arrives, using the same method that carried the original reading, including a working return path over light for Li-Fi
- A Raspberry Pi 4 pipeline that aggregates the data, adds timestamps, calculates benchmark results, and publishes them through MQTT

- A small web server on the Raspberry Pi Pico 2 W, reachable from a phone browser, showing the current reading directly from the device for independent real time verification against the app
- A Kotlin Multiplatform application with shared logic and state handling, plus native charts and native notifications on each platform
- A benchmark data set and its analysis, covering throughput, range, reliability, latency, and power use measured with INA219 sensors, along with results under interference from radio congestion, ambient light, and a blocked light path
- A comparison between the internal temperature rise and the INA219 power readings for each transmission method, tested one method at a time so the temperature rise can be attributed to that method alone
- A final report with the source code, the firmware, the circuit schematics, and the full analysis
- A record of the hours spent on the project
- A presentation with a live demonstration, including a live interference test for each method

6. Estimated Components

Component	Qty	Explanation
Raspberry Pi 4	1	Sends and receives data on all three paths, including its own LED and photodiode circuit for the two-way Li-Fi link, adds timestamps, logs the data, calculates the benchmark results, sends confirmations back to the Raspberry Pi Pico 2 W on all three methods, and publishes the results through MQTT
Phone (Android, and iOS if available)	1 to 2	Runs the Kotlin Multiplatform app, so the live data and the benchmark results can be viewed and tested
Raspberry Pi Pico 2 W	1	Acts as the sensor node, reads the internal temperature sensor, sends it through Li-Fi, BLE, and Wi-Fi, and receives the confirmation back on all three methods, including through its own photodiode circuit for Li-Fi. A second board as a spare is optional, since it is cheap and protects against losing time if the working board gets damaged during soldering or testing
LED (for the Raspberry Pi Pico 2 W and for the Raspberry Pi 4)	2	Raspberry Pi Pico 2 W
Photodiode (for example BPW34)	2	One photodiode on the Raspberry Pi 4 receives the reading from the Raspberry Pi Pico 2 W, and one photodiode on the Raspberry Pi Pico 2 W receives the confirmation from the Raspberry Pi 4, each turning its light signal back into an electrical signal

Op amp (for example LM358 or TL072)	2	Builds the transimpedance amplifier stage on each photodiode circuit, which turns the small signal from the photodiode into a clear, readable voltage
Comparator (for example LM393)	2	Turns each amplified photodiode signal into a clean digital high and low pulse, one for the Raspberry Pi 4 side and one for the Raspberry Pi Pico 2 W side
Resistor and capacitor kit	1	Sets the gain and filtering of both amplifier circuits, and supports other small circuit needs
Breadboard and jumper wires	2 to 3 sets	Used to build and test both Li-Fi circuits before soldering
INA219 current and voltage sensor	1	Measures the power used by the Raspberry Pi Pico 2 W
Small perfboard	2	Holds the final, soldered version of both Li-Fi circuits, so the connections stay stable during testing and demonstration
Enclosures	2	Protects the Raspberry Pi Pico 2 W circuit and the Raspberry Pi 4 circuit and gives the project a finished look
USB cables and power supplies	as needed	Power the Raspberry Pi Pico 2 W boards and the Raspberry Pi 4

7. Estimated Hour Breakdown

Activity	Hours
Literature review on Li-Fi, BLE, and Wi-Fi comparisons, on OOK modulation, and on benchmark methods	20
Two-way Li-Fi link, including the LED, photodiode, amplifier, and comparator circuit on both the Raspberry Pi Pico 2 W side and the Raspberry Pi 4 side, plus the firmware and logic for both directions	65
Raspberry Pi Pico 2 W BLE and Wi-Fi firmware, including sending data and receiving the confirmation	20
Pi4 pipeline for aggregation, logging, benchmarking, sending confirmations on all three methods, and MQTT publishing	40
Power measurement setup with three INA219 sensors, plus isolated, one method at a time internal temperature comparison	20
Kotlin Multiplatform app: data models, MQTT client, state handling, and UI structure	45
Native chart integration and native notifications per platform	15
Raspberry Pi Pico 2 W web server for local, real-time verification of the current reading	10
Structured benchmark testing across range, interference, and power, with repeated trials	20
Report writing, analysis, reflection, and presentation	25
Total	280